pH MEASUREMENT

INTRODUCTION

The measurement and control of pH are applied in an increasing number of fields. A classification is given below outlining some of these areas.

1. Preparations where the effiency of the process is governed by the pH of the solutions involved.

- Quality Control, in processes where maintenance of product uniformity is dependent on pH control.

- Neutralization of effluents, where the tolerated pH value of the discharge into rivers and sewers is controlled by legislation. A typical control limit is 6-9 pH.

- Corrosion inhibition, by maintenance of the pH at a suitable level.

The pH of a solution is a measure of the acidity or alkalinity of that solution. When an acid is dissolved in water the acid molecules dissociate and liberate positively charged hydrogen ions into the solution. It is these hydrogen ions that confer the acidic properties on the solution. A base, on the other hand, liberates negatively charged hydroxyl ions and it is the hydroxyl ions that confer the alkaline properties on the solution. If a solution has equal numbers of hydrogen and hydroxyl ions, like water, then the solution is neutral and is defined as being at 7 pH. When the number of hydrogen ions exceeds the number of hydroxyl ions the solution is acidic and has a pH value in the range 0-7 pH; when the number of hydroxyl ions exceeds the number of hydrogen ions the solution is basic and has a pH value in the range 7-14 pH.

It should be apparent that if the quantities of hydrochloric and acetic acid are added to water to yield two solutions of equal molar concentrations, the actual hydronium ion concentrations will be quite different for the two solutions. Yet it is not just the concentration of hydronium ions that determines the pH of a solution, but rather the concentration of active hydronium ions. Activity is affected by such things as concentration, temperature and other ions. The pH of an aqueous solution is defined as the negative logarithm of the active hydronium ion concentration or;

$$pH = -\log_{10} [\alpha H_3O^+]$$

The pH scale covers a range of 0.0 to 14.0, although it is possible to have pH values slightly beyond those limits.

Example

Calculate the amount of hydrogen ions present in a mixture with a pH of 6

$$pH = -\log_{10} [\alpha H_3O^+]$$

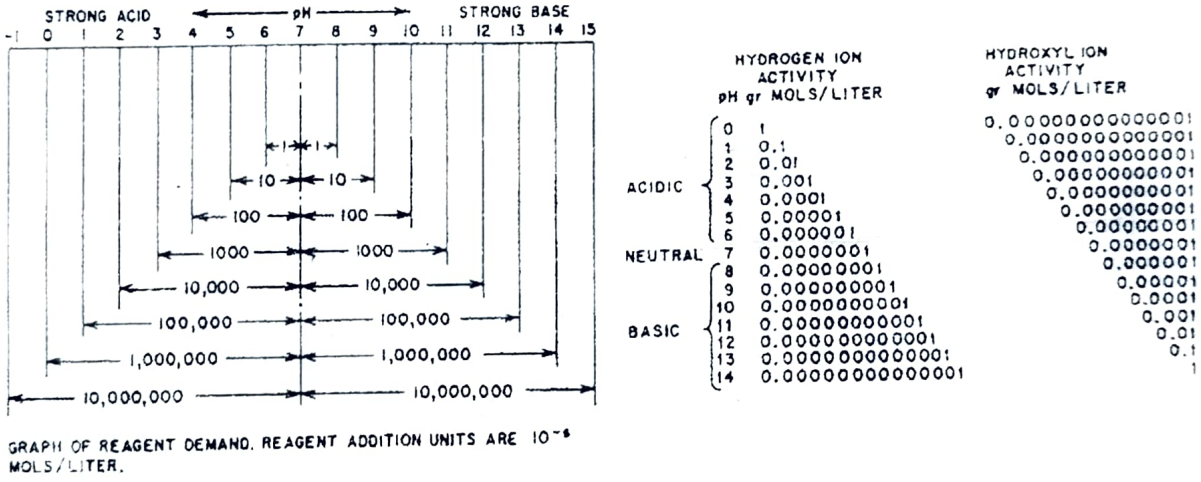
$$6 = -\log_{10} [\alpha H_3O]$$

$$6 = -\log_{10} [9]$$

$$9 - \text{anitlog } [-6]$$

$$9 = 1 \times 10^{-6}$$

There are 1×10^{-6} mole/ℓ hydrogen ions present in the solution.



GRAPH OF REAGENT DEMAND. REAGENT ADDITION UNITS ARE 10^{-6} MOLS/LITER.

FIG *The logarithmic nature of pH*

MEASUREMENT OF pH

Although it is possible to measure pH by colour change of certain pH sensitive dyes, by the use of certain meta oxide electrodes and by other techniques, by far the most widely used method employs a pH-sensitive glass membrane electrode, a reference electrode and a high input impedance potentiometer. A process pH system also will generally incorporate automatic temperature compensation and hardware for the installation of the electrode system into the process stream.

CHEMICAL BASIS FOR OXIDATION-REDUCTION POTENTIAL MEASUREMENT

The general term oxidation comes from the early recognition of combustion as the combination of combustible materials with oxygen. It is now known that when a substance such as iron combines with oxygen.

Electrons are transferred from the ion to the oxygen. Such reactions, involving the exchange of electrons, are called oxidation-reduction reactions. When electrons are added, it is reduced. The substance giving up the electrons is a reductant, or reducing agent. For example, ferrous iron is oxidized to ferric iron with the loss of one electron.

The general form of this equation would be written

$$\text{Reductant} = \text{oxidant} + ne^-$$

where n is the number of electrons. The reductant on being oxidized becomes an oxidant. The reaction may process in either direction depending on the relative electron affinity and the concentrations of the components in the system. Since free electrons cannot ordinarily exist in solutions half cells, such a reaction can take place only if a second oxidation-reduction system is present to take up the electrons.

Thus, reaction takes place between an oxidant and a reductant to produce a new oxidant and a new reductant:

$$Ox_1 + red_2 = red_1 + ox_2$$

Reaction (electron transfer) will take place directly between them and will proceed until an equilibrium is reached, i.e. until the rates of reaction in the forward and reverse directions are equal. The equilibrium state will depend upon the relative electron-holding power of the various ions and upon their relation concentrations.

The relative strength of oxidizing and reducing agents may be determined by bringing such agents together and then determining to what extent reaction occurs between them. Strength or the relative electron affinities of the agents can also be determined without direct mixing. The materials will react, even though physically separated if a means is provided for electron transfer between the materials. It has been shown that a definite electrical potential exists between such materials thus separated and that the magnitude of the potential is proportional to the relative strength of the reducing and oxidizing agents. If current is permitted to flow between separated agents, the numerical product of the current and the voltage drop (power) can be directly related to the thermodynamic change in free energy accompanying the chemical reaction.

For example, the oxidation of ferrous ion by stannic ion may be carried out by separating the iron solution from the tin solution using a semipermeable membrane and by dipping nonreactive metal electrodes in the solutions. When these electrodes are connected externally by a conductor, electrons will be given up by ferrous ions at the anode and will flow through the conductor to the cathode, where they will be taken up by stannic ions. If the reaction is allowed to proceed, the same equilibrium state will be reached as in

the previous example when solutions were mixed. If, however, a voltage is applied in the external circuit to oppose the flow of electrons, the reaction will cease when the applied voltage equals the voltage of the electrochemical cell. This voltage is then a measure of the tendency of electrons to flow in this system and corresponds to the reduction potential of the cell.

The standard electrode potential E^0 characterized the relative oxidizing ability, or electron affinity, or the system under standard conditions with respect to hydrogen. This is the potential that would be measured in a solution of mola strength at 25°C with respect to a standard hydrogen electrode. D^0 is negative for more effective reductants than hydrogen and positive for more effective oxidants.

The measured potential in a pH system is:

$$E^0 = k - \frac{RT}{nF \log_{10} (\alpha H_3O^+)}$$

where k is some constant covering a number of factors. In pH measurement absolute potentials are meaningless the change in potential per concentration unit being the important parameter in the above equation. R/nF is a constant where

R = the gas constant
F = the Faraday
n = 1
and H_3O = hydronium ion

In the foregoing discussion reference has been made to a single half-cell. Actually, solution potential cannot be measured for a single half-cell. Only the difference in voltage for two half-cells E^0 is measured between a reference electrode, such as a hydrogen gas electrode or calomel electrode that does not enter into reaction with the system and an inert electrode, such as platinum or gold or of a metal whose ions are to be determined.

SOLUTION POTENTIAL MEASUREMENTS (Oxidation-Reduction)

Solution potential measurements afford a simple and direct method of determining the relative strength of oxidizing or reducing agents in chemical systems.

There are two measurements of interest in oxidation-reduction systems:

1. Reduction potential, determined by measuring the voltage of a cell, immersed in the solution, consisting of a suitable indicator electrode and a reference electrode, the potential of the reference electrode with respect to the

standard hydrogen electrode is subtracted from or added to the measured voltage.

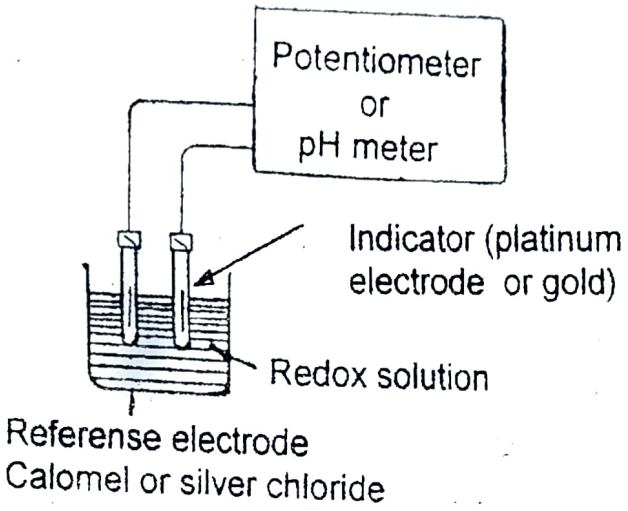
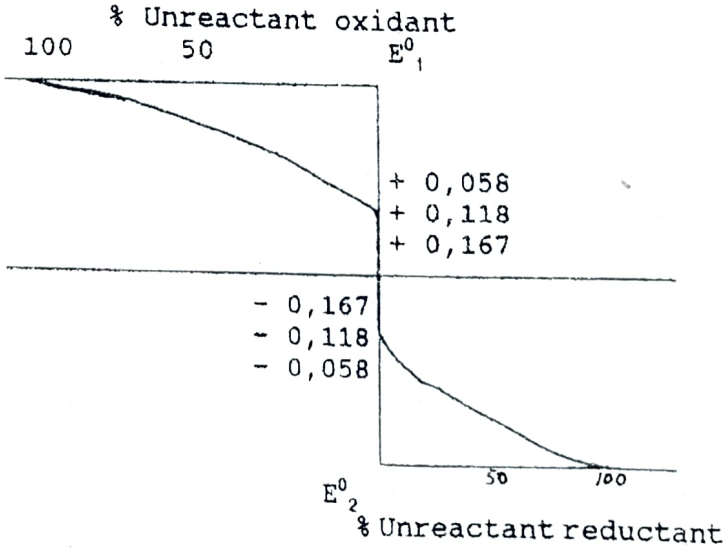


FIG *Reduction potential measurement*

2. Amount of oxidant or reductant present, determined by titration. The oxidant and reductant concentrations appear as a ratio of quantities and when this ration changes by a factor of 10, the electrode voltage changes 0,0591 volt. Electrode voltage is independent of quantity of oxidant or reductant and is determined by concentration ratio.



Potentiometric titration curve the actual equivalence voltage is determined by the equilibrium constants of the particular oxidant and reductant used,

Since various substances in solution may be either oxidants or reductants with respect to hydrogen, solution potential measurements must recognize "sense" or polarity. Although confusing conventions are in common use, the preferable system involves reduction potentials whose polarity is negative for agents that are more effective reductants than hydrogen and positive in sign for agents that are more effective oxidants than hydrogen.

By combining solution potential measurements with titrate step, it is possible to determine both the identity and concentration of a given oxidant or reductant in solution.

ELECTRODES

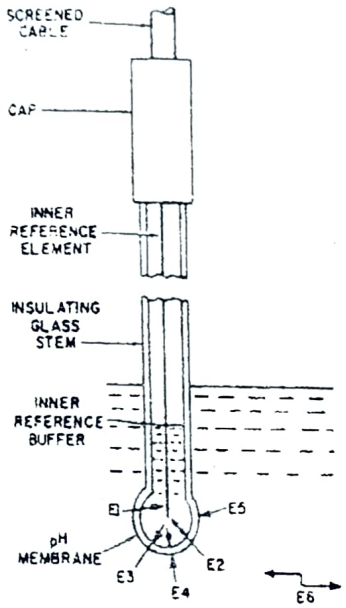
Consider initially what happens when a metallic electrode is placed in a solution. There is a tendency for positive ions of the metal to enter the solution resulting in the solution becoming positively charged. But if the solution already contains ions of the metal there is a tendency for ions to be deposited on the electrode, giving a positive charge. The electrode ultimately reaches an equilibrium potential relative to the solution. The value and sign of the potential depend on the concentration of metallic ions in the solution.

Two common examples are zinc, which assumes a negative potential relative to solutions of its salts, and copper which reaches a positive potential relative to solutions of its salts. In passing it may be noted that these two electrodes and solutions of their salts form the basis of a Daniell cell.

Hydrogen, although a gas, can be made to form the basis of an electrode.

GLASS MEMBRANE ELECTRODE

The glass electrode in generalized form consists of a pH-sensitive glass membrane fused to a glass tube. A silver wire coated with silver chloride protrudes into the thin-walled glass bulb. The bulb contains a internal filling solution, generally a buffered solution containing chloride ions.



pH Glass membrane electrode

For a glass membrane electrode to function, both surfaces of the membrane must be hydrated. Hydration occurs by absorption of water by the membrane interface in contact with the solution. There is also an exchange of univalent cations of the glass for hydrogen ions from the solution. At both surfaces of the membrane there will be a boundary potential that will be a function of the hydronium ion activity in the solution at the interface. Because hydronium ion activity for the internal filling solution is constant, the potential will be a function of the hydronium ion activity of the external solution. This potential is:

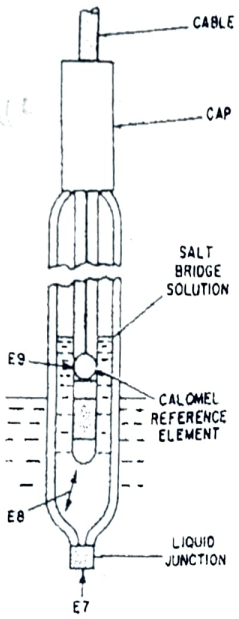
$$E = k - 0{,}059 \log [\alpha H_3O^+] \text{ at } 25°C$$

where k includes such factors as the activity of hydronium ions in the internal solution and the asymmetry potential of the membrane. Asymmetry potential arises from differences in the characteristics of the two surfaces of the glass membrane and results from such things as mechanical stresses, deterioration of the external surface by dissolution of the glass, chemical attack on the glass, dirty surfaces and so on. Thus, the potential E will not be the same from electrode to electrode or for a given electrode with relation to time for a constant concentration of active hydrogen ions. For this and other reasons, a pH system must be standardized or adjusted periodically for its entire useful life.

REFERENCE ELECTRODES

The function of a reference electrode is to complete the electrical circuit with the glass membrane electrode. It must provide a stable potential that is relatively invariable, despite changes in either the chemical composition or the physical properties of the process stream.

We will discuss a reference electrode most widely used namely the calomel electrode.



A typical calomel reference electrode

A inner tube contains calomel (mercurous chloride), potassium chloride paste and mercury. For electrical connection purposes, a platinum wire extends down the tube and makes contact with the mercury. An outer tube contains a saturated solution of potassium chloride and terminates in a porous plug. The latter can be of glass, ceramic material or asbestos fibre. The potassium chloride acts as a "salt bridge" and seeps or diffuses through the porous plug to make an effective liquid junction with the solution whose pH is to be determined.

Alternative designs have a small hole near the base of the outer tube and this is covered with a ground glass sleeve or a silicone rubber sleeve. These sleeves permit a faster rate of diffusion than with the asbestos fibre plug pattern generally. On a broad basis, the sleeve pattern is suitable for solutions of high viscosity or turbidity and with slurries, oil emulsions and soap solutions. It may be necessary after each measurement to lessen the sleeve and flush the tip of the electrode.

The choice of potassium chloride solution in the electrode is made for several reasons, but the main one is that the potential between this substance and many other solutions is extremely small and can be made very stable.

A reference electrode must be substantially stable with regard to EMF. At room temperature, the calomel electrode is reasonably satisfactory in this direction.

Duel Electrodes

Under certain conditions it may be advantages to use a dual electrode, that is, a combined glass membrane and reference electrode. This type of electrode is eminently suited to applications where the temperature difference between the two electrodes must be reduced to a minimum. Such electrodes can have an accuracy of ± 0,1 pH for a temperature variation of ± 20°F within the range 4-10 pH. Dual electrodes also have the advantage that only one aperture is required instead of two.

Calomel Electrode

The calomel electrode has been made by bringing mercury into contact with a solution saturated with mercurous chloride and containing potassium chloride at a fixed concentration.

Best results are obtained when mercurous chloride and mercury are mixed to a paste with Kcl solution and placed in contact with liquid mercury. A platinum wire immersed in the mercury is used as a lead wire.

Indicator Electrodes (Platinum - Bright)

Platinum-wire electrodes are suitable for general application as indicator electrodes.

These electrodes have been used to indicate the voltage of oxidation-reduction couples (such as the ferrous, ferric couple) and to form the conducted contact for a gas or liquid electrodes (such as the hydrogen gas electrode or bromine electrode). Platinum is, however, catalyst for some reactions and will anomalous voltage readings when used as an electrode in such systems.

Hydrogen Electrode

The hydrogen electrode is formed by immersing a portion of a platinum electrode, preferably coated with a platinum black, in a solution containing hydrogen ions and maintaining an atmosphere of hydrogen gas in contact with the exposed portion of the electrode.

This electrode is responsive to changes in hydrogen-ion concentration in the solution. When the pressure of hydrogen gas is maintained at 1 atm and the hydrogen-ion concentration at 1 mole per litre, the electrode becomes a standard hydrogen electrode (see reference electrodes, pH).

Iodine Electrodes

The iodine electrode is formed by immersing a platinum or gold electrode in a solution saturated with iodine.

Silver-Silver Chloride Electrode

This electrode has been growing in popularity in recent years because of its good stability, particularly at higher temperatures. Electrodes may be prepared by electrolyzing silver wire in a solution of hydrochloric acid with the wire as anode. Silver ions are formed at the metal surface and react with the chloride to form an adherent coat of silver chloride on the electrode. This electrode immersed in a solution of potassium chloride develops a stable voltage. It is important that the electrode wire be free from strains and that precipitated chloride by thoroughly free from electrolyte. When these precautions are observed, the silver-silver chloride electrode may be used as a primary standard.

Two-phase Electrodes

Gases such as hydrogen and oxygen, liquids such as bromine, and solids such as iodine dissolve in water to reach an exchange equilibrium with ions already in solution, producing the characteristic solution potential across the half-cell so formed. Electrodes may therefore by produced employing the gas, liquid or solid as the responsive element by making suitable contact with the solution.

Quinhydrone Electrode

The compound quinone may be electrolytically reduced to hydroquinone. The voltage of this couple is very reproducible. These compounds form a stable equimole compound called quinhydrone which dissolves to give equal concentrations of quinone and hydroquinone.

It is also very useful as a reference electrode in calibrating pH meters for the measurement of oxidation-reduction potentials A quinhydrone electrode, however, cannot be used in solution where the pH value is above 8,5.

Tungsten Electrode

The tungsten electrode has been found very useful as a reference electrode in oxidation-reduction titrations where a platinum or other noble metal indicator electrode is used. The voltage of a tungsten electrode remains relatively constant through the end point of a titrate using strong oxidizing reagents, while the voltage of the indicator electrode changes sharply at the end point.

The tungsten electrode is one very useful as an indicator electrode in measuring oxidation potentials, since its voltage usually drifts slowly over a period of contact with electrolyte and it must therefore be frequently calibrated against a calomel electrode or its equivalent as a primary standard.

MATERIALS

All materials likely to come into contact with the process liquid must be chemically resistant at all measurement temperatures. This applies both to the assembly and the sealing materials. The ideal housing will have resistance to corrosion, strength against thermal and mechanical shock and in addition, for measurement at high temperatures, impermeability to water vapour. Because permeability increases with temperature, water vapour will destroy the electrical insulation.

Both metals and plastics are used. The exact choice of materials depends upon the characteristics of the process liquid and the conditions under which the measurement is made. Metals are strong and impermeable but have limited resistance to corrosion, while offer resistance to corrosion but have an indifferent water vapour permeability.

Metals

Many metals, especially soft metals and amalgams, are specific electrodes for the measurement of the concentration of their ions in a solution. Silver, copper, mercury, antimony and others have been used. In general, metal electrode may be made by simply mounting wire rod or plate electrodes in suitable insulating mounts. A portio of the metal is immersed in the solution. Mercury is usually placed in a well or cup so that its upper surface contacts the solution. Contact with the pool of mercury is made by means of a platinum wire usually entering from the bottom. Platinum wire or foil may be electroplated with mercury to produce useful electrodes. These require frequent replacing in order to ensure reliable voltage measurements.

Platinum (Surface Active)

Platinum black is often deposited on platinum electrodes to increase surface area where the electrode is used with a gas as in the hydrogen electrode. Voltage stability is improved by this procedure. Platinum black may be deposited by electrolysis from a 3 percent solution of chlorplatic acid. Electrodes must be thoroughly rinsed after treatment. Last traces of chlorine may be removed by the generation of hydrogen on the electrode. Platinized electrodes are sometimes heated to redness to convert the black to a grey deposit and reduce catalytic activity.

Gold

Gold-wire or -foil electrodes are also suitable for general application as indicator electrodes and are to be preferred over platinum for some systems where platinum is catalytically active.

SUSPENSIONS AND DEPOSITING CONDITIONS

The problems arise when the nature of the solution interferes with the measuring function of the electrodes. Where the liquid junction device is in contact with a highly charged sediment, a spurious lowering of the pH can occur. Serious potential effects are rarely involved in measurements on the dispersed suspension and the main precaution necessary is to prevent blockage of the liquid junction of the reference electrode. Sols may also give rise to residual junction potentials and gels sometimes cause sluggish glass electrode response. Sols and suspensions normally cause difficulties only if they are allowed to settle. Constant agitation of the process bath prevents this, and a reservoir-fed reference electrode minimized the flow of particles into the junction.

Although coagulated solids do not usually cause glass electrode problems, their presence and the formation of colloids can present serious difficulties. The material deposits on the glass electrode membrane, and depending on the nature of the precipitate and the temperature of the solution, inhibits the electrode response.

Film formation on the glass electrode surface is the principal problem. Where oils are dispersed in water, the globules may coagulate on the glass and inhibit its response. Where the measurement is on viscous materials such as glue or latex sluggishness is generally experienced. This is partly due to lack of water and glass electrodes requiring less hydration for their functioning are generally more satisfactory.

The most serious problem is loss of response due to adherence of dispersed oil on the membrane. This is more likely to occur with fine emulsions than with coarse ones where the oil tends to separate out. When measuring slowly flowing coarse emulsions, the electrodes should be situated high in the solution of dense oils and low in the solution for oils that float on water.

WAYS OF REDUCING THE DEPOSITION RATE

1. Careful choice of locating the electrode assembly, with a compromise between the greatest distance from the precipitation region and the desirability of close proximity to minimize process control lags.

2. To incorporate the assembly into the liquid circulating pipeline from the deposition tank.

3,. Bleed arrangement which is used on effluent treatment plants.

4. Flow systems can be utilized in cases where finely divided particles can exert a cleaning action.

A number of mechanical devices can also be used to keep the electrodes clean, but eventually solvent cleaning is necessary for all types of deposit.

ELECTRODE CLEANING AND CLEANING DEVICES

In order to maintain their speed of response and accuracy of determination, the electrodes should be cleaned periodically. The frequency of cleaning and the method which should be applied are dependent on the nature of the process liquid.

In general, washing with water and wiping with cotton wool soaked in dilute hydrochloric acid will suffice, but in severe cases the electrodes may require soaking in solvent. Inorganic solvents are preferable for removing greasy and oily deposits. Alcoholic and acetone are commonly used but care must be taken that another deposit just as immiscible is not on the electrode. Organic solvents have a dehydrating effect on the membrane and only brief immersion in the solvent followed by soaking in hydrochloric acid should be employed.

Electrode cleaning frequency may be considerably reduced by the fitting of an in site cleaning device. One such device provides for a met of solvent to be periodically directed at the electrodes. Another technique, providing continuous cleaning without the introduction of solvents, employs an ultrasonic transducer attached to a thin plate and operated by a remotely mounted generator.

Leads

Screening is necessary for the insulated cable attached to the glass electrode. Polythene-insulated standard wire enclosed in metal braid is normally used with an outer sheath of PVC. At temperatures above 140°F, the hot solution itself acts as a screen for the immersed cable, but a covering of polypropylene or Teflon is used to insulate the cable core.

The length of the lead is limited by its capacitance, for this governs the reading response of the measuring instrument. The actual length that can be tolerated depends upon the capacity of the instrument being used and on the glass electrode resistance.

Bridges

It is often advisable to separate the two half-cell of a system to be studied and still maintain electrical contact. The salt bridge is used for this purpose. Such a bridge is

positioned to dip into each electrolyte and provide a conductive path with very slight diffusion of the electrolyte into either half-cell.

Important design criteria for salt bridges are:

1. A positive pressure between the salt solution in the bridge and the solution outside the bridge so that the direction of flow opposes diffusion into the bridge. Contaminants allowed to reach the half-cell through the bridge usually lead to unreliable measurements.

2. A barrier at the termination of the bridge such as a capillary, agar plug, round glass stopper, etc. provides high flow velocities at small volumes to avoid excessive dilution and contamination of solutions being measured.

This arrangement is effective in isolating the solution from contamination, reducing liquid junction potentials, and in isolating solutions where the temperature of the sample solution may be different from that of the reference electrodes.

MEASURING CIRCUITS

The emf from the pH electrode system is d.c. and the measuring instruments fall into two broad classes:

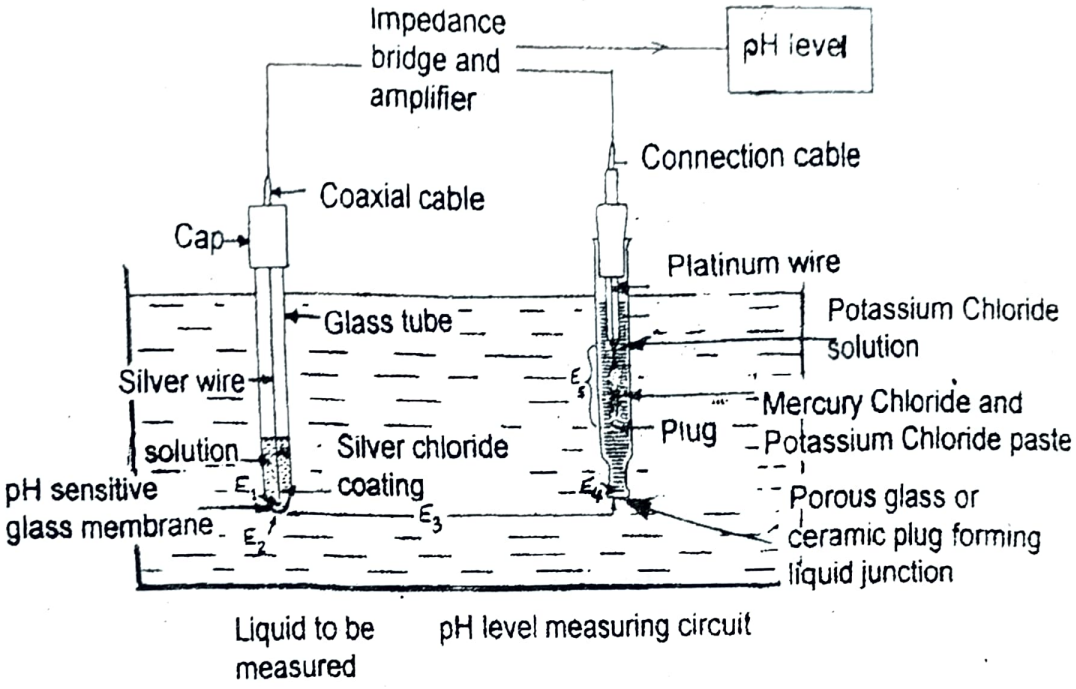
1. Those which amplify directly the electrode emf so that the output can operate a standard indicator, recorder and controller. This class involved a dc amplifier with very stable characteristics.

2. Those which compare the electrode emf with a known standard emf. The self-balancing potentiometer is the obvious instrument here, but it may involve the interposition of some form of dc-ac converter unit which transposes the dc emf signals to corresponding ac ones.

Whichever class is used, the following general requirements should be met:

1. The instrument to be substantially free from errors caused by supply voltage or frequency variations.
2. The zero reading to be stable over long periods.
3. The current taken from the electrode system to be extremely small to avoid polarization effects.
4. The design to include compensation for variation in characteristics of components or be such that the output is independent of such variations.
5. Temperature compensation to be included, either manual or automatic.
6. Facilities to be included for easy checking of calibrations.

TYPICAL MEASURING CIRCUIT FOR pH CONTENT OF A LIQUID



pH Level measuring circuit

The circuit consist of two electrodes placed in a liquid of which pH level we determine. To complete the circuit the liquid must be of conductive substance and a impedance bridge is used to determine the voltage drop between the two electrodes.

A stable voltage E_1) will be induced by the internal solution of the glass membrane electrode. This will be a known constant value. Due to the ionization taking place at the pH sensitive glass membrane caused by the amount of hydronium ions present another voltage E_2 will be induced.

If the reference voltage induced by the calomel electrode, E_4 and E_5, differs from the value at E_2 an potential difference, E_3, will exist between the two electrodes. This potential difference, E_3, will be increased or decreased by the amount of hydronium ions present at the glass membrane.

A impedance bridge is used to determine this potential difference, E_3, and amplify the signal to be send to a measuring instrument calibrate in a pH scale.

POTENTIAL MEASURING INSTRUMENTS

Potentiometer-Galvanometer Systems.

The potentiometer-galvanometer system affords a simple means of measuring oxidation-reduction potentials.

It is essential that measurement be made with no appreciable withdrawal of current in order to avoid polarization at the electrodes.

Vacuum-tube Voltmeter.

The mosfet voltmeter as a pH meter is available in many commercial models. These instruments incorporate potentiometer and galvanometer circuits and in some models can be switched from pH scale readings to millivolt scale readings. (Oxidation-reduction potentials are measured on the millivolt scale: pH meters for Redex measurement.) Measurements of oxidation-reduction potential is a very simple operation with the pH meter. Commercial electrodes are available from the various manufacturers for use with their instruments. Special electrodes may be made if required.

Potentiometric Titrations

The direct-reading pH meter is most convenient for the determination of titration end points. Electrode potential measurements are made as above as reagent is added to the sample and the electrode potential is plotted against the volume of reagent added. For simple titractions of univalent, strong electrolytes the end point of the titration is at the maximum slope of the curve, which is the mid-pint of the steep portion.

If a titration is carried out in such a system and at the end point the reversible reaction is no longer possible, one electrode becomes polarized and the indicator current suddenly stops flowing. This is called a "dead-stop" end point and is extremely sensitive.

Automatic Titration and Control

Potentiometric measurement of oxidation and reduction potentials, as applied both to direct measurement and to the determination of titration end points, is a procedure well adapted to automatic operation. Automatic instrumentation for titration and control of both batch-type and continuous flow processes in the field of pH is well advanced and the growth of this phase of instrumentation in the oxidation-reduction field is in progress.

Factors Affecting Accuracy of Measurement

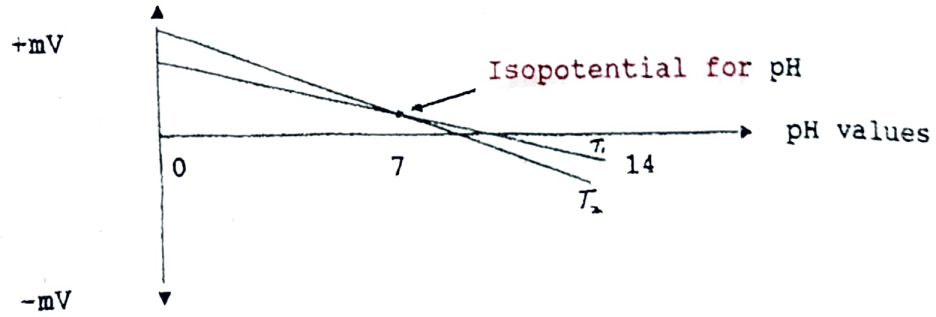
1. Temperature
2. Mixtures and reaction rates.

3. Electrode surface conditions.

Temperature

The main requirement is to correct for scale length changes, which can be accurately calibrated and compensated for by altering the resistance of the feedback circuit. This change in resistance may be effected by means of a resistance thermometer, directly measuring the sample temperature, or by means of a manual control which may be set to the expected mean sample temperature. It is also possible to compensate for zero shift by arranging that the current in the feedback circuit is zero at the isopotential value of the electrodes. This is effected by means of a variable resistor known as the isopotential control.

For certain applications where little variation is expected in the sample temperature, and where standard electrodes of a chosen isopotential value are used, it is possible to dispense with the methods of temperature compensation described above. Instead, a pre-set temperature control is used which may be set at a value about which a change of no more than ± 1 is expected. The accuracy of such a system is greatest when the pH value being measured is close to the isopotential value of the electrodes used, but should not be greater than 0,1 pH at any point over the whole range of 0-14 pH.



Variation of electrode emf with temperature

It can be calculated that potential changes at a rate of 0,186 mv/°C. The slope mv/pH for several temperatures is given below:

Temp °C	*mv/pH*
0	54,20
25	59,14
50	64,08
75	69,02
100	74,00

Commercial pH analyzers generally provide manual or automatic temperature compensation or both. In general, this is done by changing the gain or sensitivity of the instrument. For example, an automatic temperature compensator consists of an element in contact with the process stream which changes in resistance with temperature. At the isopotential point the electrodes will be self compensating.

A manual thermocompensator is calibrated variable resistor. It must be pointed out that aqueous solutions, even buffers solutions, change in pH with temperature. This is especially true of highly alkaline solutions that exhibit quite dramatic changes in pH with temperature, whereas acid solutions change only to a minor degree. These changes are very real and no attempt is made, or should be made, to affect them with thermocompensation.

MIXTURE AND REACTION RATES

The combination of the oxidized and reduced form of a given substance is called an oxidation-reduction couple. When two or more such couples are present in the same solution, a reaction must take place between them until equilibrium is reached. At equilibrium the oxidation potential calculated for the solution from the ion concentrations using the Nernst equation will be the same for either ion couples. The infinitely high voltage called for by the Nernst equation when the ratio of the ion concentrations approached infinity is obviously never observed in actual solutions. Substantially pure oxidizing agents, for example, are always able to react with minute concentrations of impurities or with water to produce small amounts of the reduced form of the couple. Drifting electrode voltages are obtained only when the concentration of the ion present in smaller amount reaches a level of about 10^{-5} mole per litre. In most practical systems one ion is usually reduced to a concentration below this limit. The oxidation potential is determined by the other ion couple. Thus in the calculation of titration end-points potentials, data for ion concentrations of one couple are used on one side of the end point and data for the other couple on the other side.

When oxidizing or reducing agents which react slowly are present in solution, electrode potentials are unstable and sometimes drift slowly over long periods of time. Solutions or reducing agents which are exposed to oxygen often show this behaviour.

ELECTRODE-SURFACE CONDITIONS

Metallic electrodes are subject to poisoning by trace quantities of impurities in electrolytic solutions and in some cases by exposure to the atmosphere. Precautions must therefore be taken in preparing or storing electrodes and in following procedures developed for producing reproducible surface conditions in order to obtain reliable readings on various systems. Treatment with strong oxidizing agents such as hot chromic acid or nitric acid solution has been found effective for preparing platinum, gold and

tungsten electrodes for use. Thorough rinsing is required following treatment. Cleaning with fine abrasive is sometimes required for metal electrodes. Electrode activity is generally more stable when electrodes are stored under water.

Some difficulties in making precise measurements with glass-electrode systems are encountered because of electrical leakage due to moisture on the electrode bodies. The use of silicone coatings on the electrodes - particularly glass electrodes - has been found effective in eliminating this source of trouble.

AMPLIFIERS

The relatively high electrical resistance of glass electrodes imposes stringent design requirements on pH meters, since without amplification, satisfactory discrimination will not be achieved in the measurement. Also, the measurement must be made in open-circuit conditions to avoid polarization of the electrodes; therefore an amplifier unit of very low and stable leakage current is required.

Direct Coupled Amplifiers

The first stage in the amplification circuit consists of a semi-electrometer valve with a grid current of less than 10^{-11} amperes. The potential difference between the electrodes is applied to his valve, the potential of the glass electrode going to the grid. Careful stabilization of the internal power supplies is essential to care for the wide variation in external supplies which may occur.

A well-designed pH meter using this type of amplifier should maintain a zero stability within 0,02 pH for periods of more than 24 hours.

Vibrating Condenser and Contact Modulated Amplifiers

Both of these types use on input system which converts the DC signal from the glass electrode into AC permitting the use of more stable AC amplifiers in the succeeding stages, in the vibrating condenser amplifier, the electrode potential difference is applied to the fixed plate of the condenser while the other plate is driven by a solenoid to produce a change in capacitance at a fixed gas-filled and sealed to prevent deterioration. The zero stability of pH meter using a vibrating condenser amplifier will remain within 0,005pH for periods or more than 24 hours.

In the contact modulated or low frequency chopper amplifier, a high grade condenser is alternately charged by the electrodes and switched every second to the input of a self-balancing potentiometer via an amplifier. If a change in pH occurs, an amplified pulse is fed to the galvanometer and the slide wire adjust itself until the potential across the slider is equal to the voltage presented by the condenser. This arrangement has the

advantages that the input resistance is solely limited by the physical insulation of the capacitor and switch mechanism, and the zero stability is independent of the input amplifier and is governed only by the accuracy of the self-balancing potentiometer.

Receivers (any receiver can be connected to the pH measure)

pH meters are built in various configurations to suit particular industrial requirements. Most instruments are designed for either wall or panel cases, which may be hosed down without risk of damage. Accessibility is important of servicing, and instruments are designed so that they may easily be withdrawn form their cases.

Increasing use is being made of modular systems, particularly where the measurement of pH is incorporated as part of a continuous process. In such cases a pH amplifier, and indicating meter and one or more control conduces are assembled in an industrial instrument case.

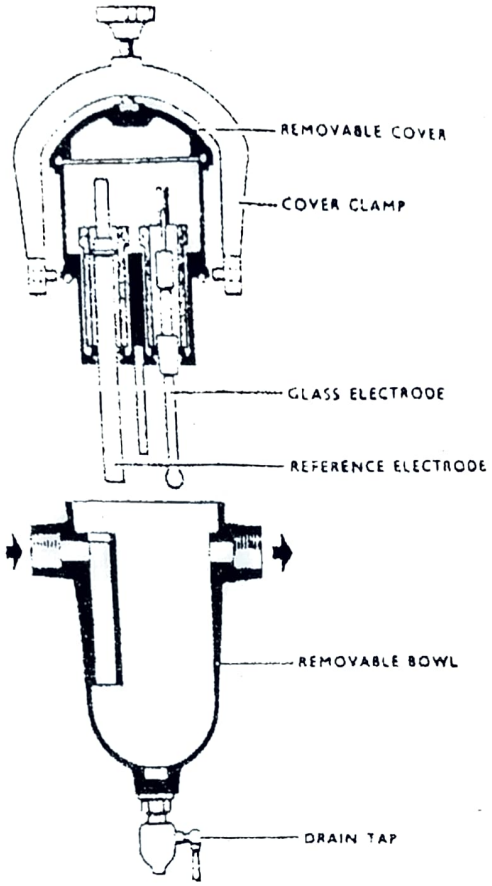
Depending on local requirements, a pH measuring system may incorporate a recorder in addition to, or instead of, an indicating meter. Monitoring of pH at a number of pints may be carried out using a single measuring instrument in conjunction with an electrode switching box and a multi-pint recorder. The switching box connects the electrodes to the measuring instrument in synchronism with the stepping mechanic in the recorder. Simple two-way manual switching boxed are also available.

When measurements are to be made in hazardous areas, equipment certified as intrinsically safe must be used. If this is the case, then the electrode assembly is incapable of seeing sufficient energy to cause ignition and therefore only the readout device requires explosion proof housing or being mounted remotely in a non-hazardous area.

ELECTRODE SYSTEMS FOR INDUSTRIAL PROCESSES

It will be realized that the electrodes shown in their simple state may not be suitable for industrial processes without further protection and it is proposed to give brief details of two typical industrial electrode systems, the continuous flow and total immersion type.

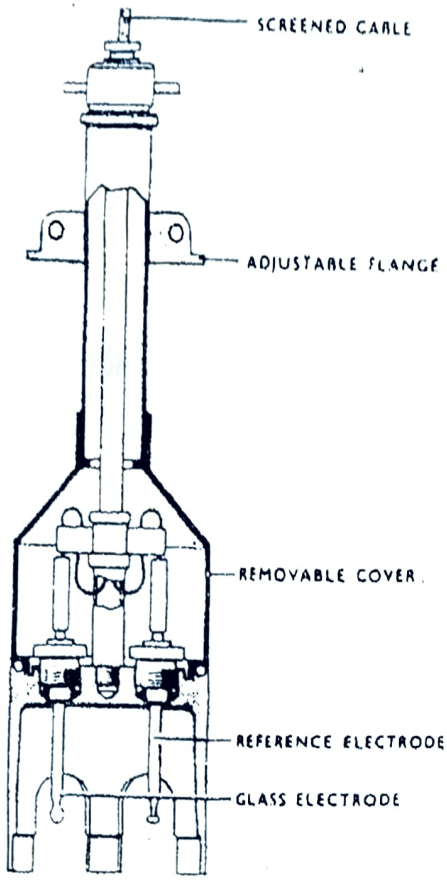
THE CONTINUOUS FLOW TYPE



Flow type pH electrode assembly

In it the measuring glass electrode and the reference electrode are fixed in a removable cover assembly, but dip down into the lower bowl when in use. The process liquid flows from left to right round the electrodes.

THE TOTAL IMMERSION TYPE (DIPPING TYPE)



Dip type electrode assembly

The measuring glass electrode and the reference electrode are fixed to the assembly with a cover to allow excess to electrodes. The assembly will then be totally immersed into a tank.

APPLICATIONS OF SOLUTION POTENTIAL MEASUREMENT

Solution potential measurements have important applications in such widely varied chemical processed as biological systems and dyestuffs, sewage treatment water purification and chemical process determination of water in organic materials by the Karl Fischer titration. Another recently developed application of considerably importance is the determination of chloride ion. There are not simple rules by which any possible application of solution potential principles can be tested because of the wide variety of electrode systems and such practical considerations as electrode maintenance.